VIDYAVAHINI GROUP OF INSTITUTIONS
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PROJECT REPORT ON
MINI PROJECT

AI Virtual Hand Cursor Using Hand Gesture Recognition

Touch-Free Human-Computer Interaction Powered by Computer Vision & MediaPipe

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1. Abstract & Introduction

The **AI Virtual Hand Cursor** is an artificial intelligence and computer vision-based system engineered to control operating system mouse operations using touch-free hand gestures. Standard computer setups require continuous physical contact with peripherals such as a mouse or trackpad, which can cause fatigue, hardware breakdown, or inconvenience in touch-free environments. This project captures video via a webcam and processes frames in real-time utilizing Python, OpenCV, and MediaPipe. A 21-point hand landmark estimation pipeline detects spatial finger configurations and calculates precise gestures. These are then mapped directly to virtual cursor navigation, left-click, right-click, and scrolling via operating system kernel events and PyAutoGUI.

2. Existing System vs. Proposed System

Feature / Aspect	Existing System (Physical Mouse)	Proposed System (AI Virtual Cursor)
Physical Contact	Requires constant hand contact with surface.	Completely touch-free; operates in open air.
Hardware Dependency	Requires physical mouse, mousepad, batteries/wire.	Uses existing built-in or USB webcam.
Hardware Failure	Switches fail, cables fray, optical sensors collect dust.	Zero moving physical parts; pure software.
Hygiene & Accessibility	Transmits germs; unsuitable for sterile/medical areas.	Sterile, hygienic, accessible for touch-free computing.
Portability	Extra accessory to carry and connect.	Runs directly on laptop/PC hardware.

3. Detailed Module Architecture

As specified in Section 5 of the project synopsis, the system comprises 7 functional modules:

No.	Module Name	Implementation Details & Technical Role
1	**Video Capture**	OpenCV captures live video frames via DirectShow (cv2.CAP_DSHOW) at 30 FPS, avoiding Windows Media Foundation deadlocks.
2	**Hand Detection**	MediaPipe Tasks Vision locates hands in each frame using lightweight convolutional networks optimized with CPU XNNPACK.
3	**Landmark Tracking**	Extracts 21 3D landmarks (x, y, z) per hand, tracking fingertips (Thumb 4, Index 8, Middle 12, Ring 16, Pinky 20) and joints.
4	**Gesture Recognition**	Evaluates geometric finger extension ($y_{tip} < y_{pip}$) and Euclidean pinch thresholds ($d < 38px$) to determine active commands.
5	**Cursor Control**	Maps active camera boundary coordinates to screen pixels using linear interpolation, adaptive smoothing, and tremor deadzone filtering.
6	**Mouse Action Control**	Converts gestures into left-clicks, right-clicks, and scrolling via native Win32 User32 APIs and PyAutoGUI with click-lock freezing.
7	**Visual Feedback**	Renders real-time HUD overlays: hand skeleton bones, golden active box, dynamic glowing action banners, and live FPS.

4. Mathematical Formulations & Signal Processing

A. Euclidean Distance Metric for Pinch Detection:

The spatial Euclidean distance d between landmark points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is defined as:

```
d(P_1, P_2) = sqrt((x_2 - x_1)^2 + (y_2 - y_1)^2)
```

• **Left Click Trigger**: If d(Thumb_4, Index_8) < 38 pixels.
• **Right Click Trigger**: If d(Thumb_4, Middle_12) < 38 pixels.

B. Active Frame Coordinate Linear Projection:

Camera margins $M_x = 100px, M_y = 80px$ ensure reaching monitor corners comfortably without reaching frame borders:

```
S_x = ((C_x - M_x) / (W_cam - 2*M_x)) * Screen_Width
S_y = ((C_y - M_y) / (H_cam - 2*M_y)) * Screen_Height
```

C. Speed-Adaptive Exponential Moving Average (EMA) Smoothing:

To reconcile smooth tracking with responsiveness, smoothing factor α adapts to motion velocity Δ:

```
P_t = P_{t-1} + (S_t - P_{t-1}) / alpha
```

• If velocity $\Delta > 60px$: $\alpha = 3$ (Ultra-fast responsive tracking across screen).
• If velocity $\Delta < 20px$: $\alpha = 7$ (High-damping stability over small icons/buttons).

D. Involuntary Tremor Deadzone Filter & Click-Lock Freezing:

• **Deadzone Suppression**: If $|S_t - P_{t-1}| < 8px$, the position is held constant: $P_t = P_{t-1}$.
• **Click-Lock Freezing**: The moment a pinch click occurs, coordinates are locked at target position for $0.35s$. This completely eliminates the natural involuntary finger displacement that causes target drift when clicking buttons.

5. Gesture Interaction Schema

Mode	Hand Assigned	Hand Gesture / Posture	Dispatched Mouse Operation
Dual-Hand (Default)	Right Hand	Index finger extended pointing	**Move Pointer** across desktop
Dual-Hand (Default)	Left Hand	Pinch Thumb (4) + Index (8)	**Left Click** (Locks position)
Dual-Hand (Default)	Left Hand	Pinch Thumb (4) + Middle (12)	**Right Click** (Context Menu)
Dual-Hand (Default)	Left Hand	Index + Middle fingers swipe vertical	**Scroll Up / Down**
Dual-Hand (Default)	Left Hand	Open Palm / Hand at rest	**Neutral / Hover Position**
Single-Hand (Fallback)	Either Hand	Index finger alone extended	**Move Pointer**
Single-Hand (Fallback)	Either Hand	Pinch Thumb + Index finger	**Left Click**
Single-Hand (Fallback)	Either Hand	Pinch Thumb + Middle finger	**Right Click**
Single-Hand (Fallback)	Either Hand	Index + Middle fingers extended	**Scroll Up / Down**

6. Hardware and Software Requirements

Category	Component	Specification / Minimum Requirement	Role in System
Hardware	Processor	Intel Core i3 2.0 GHz or AMD equivalent	Processes vision frames and ML tensors
Hardware	Memory (RAM)	Minimum 4 GB (8 GB Recommended)	Holds video buffer and GUI structures
Hardware	Webcam	Built-in or External USB (720p / 480p, 30 FPS)	Captures optical input frames
Hardware	Storage	Minimum 500 MB free space	Application files, model weights, dependencies
Software	Operating System	Windows 10 / Windows 11 (64-bit) or Linux	Host operating environment
Software	Programming Language	Python 3.10 – 3.14	Core development and algorithmic logic
Software	OpenCV	opencv-python >= 4.8.0	Frame capture, color conversion, HUD rendering
Software	MediaPipe	mediapipe >= 1.0.0 (Tasks Vision)	Hand detection & 21 3D landmark tracking
Software	PyAutoGUI / Win32	pyautogui >= 0.9.54 & User32 kernel	Dispatches hardware mouse events
Software	PyQt6	PyQt6 >= 6.11.0	Modern GUI desktop interface & floating HUD

7. Test Cases & Verification Results

Test ID	Target Component	Input Condition & Stimulus	Expected Output	Result
TC-01	HandDetector	Blank / Null frame (no hands)	Empty list returned safely without crash	**PASSED**
TC-02	MouseController	Camera edge coordinate mapping	Clamped within [0, Screen Width/Height]	**PASSED**
TC-03	TremorFilter	Micro-displacement < 8px	Position invariant (deadzone activated)	**PASSED**
TC-04	GestureClassifier	Dual-hand: left thumb+index pinch	Dispatches LEFT_CLICK gesture	**PASSED**
TC-05	GestureClassifier	Dual-hand: left thumb+middle pinch	Dispatches RIGHT_CLICK gesture	**PASSED**
TC-06	GestureClassifier	Dual-hand: 2 fingers vertical motion	Dispatches SCROLL_DOWN / SCROLL_UP	**PASSED**
TC-07	ClickLock	Finger shift during click event	Cursor locked; zero target drift	**PASSED**
TC-08	Visualizer HUD	Full frame overlay rendering	Correct BGR shape & skeleton rendering	**PASSED**

8. Conclusion & Future Scope

The **AI Virtual Hand Cursor** successfully demonstrates an accurate, touch-free replacement for a traditional hardware mouse. By utilizing MediaPipe's high-fidelity landmark tracking and solving human-computer interaction challenges (such as click jitter, tremor drift, and dual-hand control isolation), the system provides fluid desktop navigation across all applications. Automated test coverage validates 100% compliance with all 7 modules outlined in the BCA Mini Project synopsis.

Future Enhancements:

• Virtual drag-and-drop with pinch-and-hold gestures.
• Multi-monitor spanning and dynamic monitor switching.
• Voice command fusion (multimodal speech + gesture controls).